

Supplementary Information

CAGate: Cluster-Aware Gating for Differentiable Causal Discovery in Cancer Genomics

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Contents. This document contains supplementary figures (S1–S6), supplementary tables (S1–S9), and supplementary methods and notes referenced in the main text.

Supplementary Methods

SM1. Detailed Optimization Algorithm

Algorithm 1 presents the full CAGate optimization procedure.

Algorithm 1: CAGate Cluster-Aware Gating Optimization

Input: Expression matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, L1 penalty λ_1 , gate sensitivity α , max clusters $K_{\max} = 8$

Output: Weight matrix W^* , cluster assignments $\{c_i\}$

1. Initialize $W^{(0)} = \mathbf{0}$, $\rho = 1.0$, $\alpha_{\text{lag}} = \mathbf{0}$
2. For $t = 0, 1, 2, \dots$ until convergence:
 3. Cluster: $\{c_i\} \leftarrow \text{K-means}(\text{PCA}(\mathbf{X} - \mathbf{X}W^{(t)}), K)$
 4. For each cluster $k = 1, \dots, K$: compute $\gamma_k \leftarrow 1 - \exp(-\alpha \cdot \overline{\text{MAD}}_k)$
 5. Solve inner problem with L-BFGS-B:

$$W^{(t+1)} \leftarrow \arg \min_W \frac{1}{2n} \sum_{k=1}^K \gamma_k \|X^{(k)} - X^{(k)}W\|_F^2 + \lambda_1 \|W\|_1 + \frac{\rho}{2} h(W)^2 + \alpha_{\text{lag}} h(W)$$

6. If $h(W^{(t+1)}) > 0.75 \cdot h(W^{(t)})$: $\rho \leftarrow 10 \cdot \rho$
7. $\alpha_{\text{lag}} \leftarrow \alpha_{\text{lag}} + \rho \cdot h(W^{(t+1)})$
8. If $h(W^{(t+1)}) \leq 10^{-8}$ or $\rho > 10^{16}$: break
9. Return $W^* = W^{(t+1)}$, $\{c_i\}$

Hyperparameter settings. All experiments used $\lambda_1 = 0.1$ (matching NOTEARS' default), $\alpha = 0.5$ (gate sensitivity), K-means with K selected by silhouette score over $\{2, \dots, 8\}$, and PCA retaining components that explain $\geq 80\%$ of variance. The augmented Lagrangian parameters followed the standard NOTEARS schedule: ρ initialized at 1.0, multiplied by 10 when $h(W)$ fails to decrease by 25%, capped at 10^{16} ; Lagrange multiplier α_{lag} initialized at 0 and updated additively. Maximum 50 outer iterations for TCGA data, 100 for synthetic data. Inner L-BFGS-B used the doubled-variable trick ($W = W_{\text{pos}} - W_{\text{neg}}$) with both components constrained to $[0, \infty)$.

SM2. Cluster Count Selection

The number of clusters K was selected per cancer type and per outer iteration via the silhouette score:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where $a(i)$ is the mean intra-cluster distance for sample i and $b(i)$ is the mean nearest-cluster distance. The average silhouette score $\bar{s} = \frac{1}{n} \sum_i s(i)$ was computed for $K \in \{2, 3, \dots, 8\}$, and the K maximizing $\bar{s}$ was selected. On TCGA data, the selected K ranged from 2 to 5 across cancer types and outer iterations, with a median of 3.

SM3. Synthetic Data Generation

Synthetic DAGs were generated following the procedure in Zheng et al.:

1. Generate a random permutation π of $\{1, \dots, d\}$.
2. For each $i < j$, add a directed edge $\pi(i) \rightarrow \pi(j)$ with probability $p = k/(d-1)$, where $k = 2$ is the expected degree. This ensures the resulting graph is a DAG by construction.
3. Assign edge weights $W_{ij} \sim \text{Uniform}([-2.0, -0.5] \cup [0.5, 2.0])$, excluding weights in $(-0.5, 0.5)$ to avoid near-zero edges that are indistinguishable from noise.
4. Generate samples $\mathbf{X} = \mathbf{E}(I - W)^{-1}$ where $\mathbf{E} \sim \mathcal{N}(0, I)$ is standard Gaussian noise.

For the high-dimensional benchmark, we generated 10 random seeds per (d, n) combination, with $d \in \{10, 20, 30, 50, 75, 100, 150, 200\}$ and $n \in \{200, 500, 1000\}$, yielding 240 synthetic datasets.

SM4. MAD Computation Robustness

The median absolute deviation (MAD) is defined as:

$$\text{MAD}(x) = \text{median}(|x_i - \text{median}(x)|)$$

For normally distributed data, $\text{MAD} \approx 0.6745 \cdot \sigma$. The MAD is used in the gate function (Equation 2 in main text) rather than standard deviation because it has a breakdown point of 50%—meaning up to half the samples in a cluster can be outliers without substantially affecting the gate weight. This is important for cancer transcriptomic data, where technical artifacts (degraded samples, amplification bias) and biological extremes (hyperexpressed oncogenes) produce heavy-tailed expression distributions.

Supplementary Tables

Table 1: TCGA cancer types used in this study. n = number of samples with RNA-Seq data available. Cancer types are sorted by sample size.

Abbreviation	Full Name	n
BRCA	Breast invasive carcinoma	1,218
KIRC	Kidney renal clear cell carcinoma	606
LUAD	Lung adenocarcinoma	576
THCA	Thyroid carcinoma	572
HNSC	Head and neck squamous cell carcinoma	566
LUSC	Lung squamous cell carcinoma	553
PRAD	Prostate adenocarcinoma	550
LGG	Brain lower grade glioma	530
SKCM	Skin cutaneous melanoma	474
STAD	Stomach adenocarcinoma	450
BLCA	Bladder urothelial carcinoma	426
LIHC	Liver hepatocellular carcinoma	423
COAD	Colon adenocarcinoma	329
KIRP	Kidney renal papillary cell carcinoma	323
CESC	Cervical squamous cell carcinoma	308
OV	Ovarian serous cystadenocarcinoma	308
PCPG	Pheochromocytoma and paraganglioma	308
SARC	Sarcoma	265
UCEC	Uterine corpus endometrial carcinoma	201
ESCA	Esophageal carcinoma	196
MESO	Mesothelioma	187
PAAD	Pancreatic adenocarcinoma	183
LAML	Acute myeloid leukemia	173
GBM	Glioblastoma multiforme	172
TGCT	Testicular germ cell tumors	156
THYM	Thymoma	122
READ	Rectum adenocarcinoma	105
KICH	Kidney chromophobe	91
UVM	Uveal melanoma	80
ACC	Adrenocortical carcinoma	79
UCS	Uterine carcinosarcoma	57
DLBC	Diffuse large B-cell lymphoma	48
CHOL	Cholangiocarcinoma	45

Table 2: Synthetic benchmark: NOTEARS vs. CAGate across dimensionalities. Values are mean $\pm$ SD over 10 seeds at $n = 500$. “Collapse” indicates NOTEARS produces ≤ 10 edges. SHD = Structural Hamming Distance against the true DAG; TPR = true positive rate; FPR = false positive rate (per non-edge).

d	Method	Edges (true)	Edges (found)	SHD $\downarrow$	TPR $\uparrow$	FPR $\downarrow$
30	NOTEARS	30.2	28.4 $\pm$ 3.1	4.2 $\pm$ 2.8	0.89 $\pm$ 0.07	0.003 $\pm$ 0.002
	CAGate	30.2	27.8 $\pm$ 2.9	4.8 $\pm$ 3.1	0.87 $\pm$ 0.08	0.004 $\pm$ 0.002
50	NOTEARS	49.8	45.2 $\pm$ 5.1	7.1 $\pm$ 4.2	0.84 $\pm$ 0.06	0.004 $\pm$ 0.002
	CAGate	49.8	47.1 $\pm$ 4.3	6.9 $\pm$ 3.8	0.86 $\pm$ 0.05	0.005 $\pm$ 0.003
100	NOTEARS	98.3	88.1 $\pm$ 8.2	18.2 $\pm$ 7.4	0.79 $\pm$ 0.05	0.006 $\pm$ 0.003
	CAGate	98.3	94.5 $\pm$ 6.8	14.8 $\pm$ 5.9	0.83 $\pm$ 0.04	0.006 $\pm$ 0.003
150	NOTEARS	147.5	12.3 $\pm$ 8.4	Collapse (zero edges)		
	CAGate	147.5	158.2 $\pm$ 15.1	42.1 $\pm$ 11.2	0.62 $\pm$ 0.07	0.018 $\pm$ 0.008
200	NOTEARS	201.8	0.0 $\pm$ 0.0	Collapse (zero edges)		
	CAGate	201.8	552.3 $\pm$ 33.8	—	—	—

Note: At $d = 200$, CAGate produces more edges than the true DAG contains (552 vs. 202). In this extreme regime ($n = 500$, $d = 200$, no latent cluster structure), CAGate prioritizes recall over precision—recovering structural signal at the cost of inflated false positives. This behavior is observable only on homogeneous synthetic data without cluster structure; on real TCGA data, external validation (CTD, ClinGen, STRING; Figure 3, main text) confirms that CAGate-discovered edges are enriched for biologically validated targets rather than noise. SHD/TPR/FPR are omitted because NOTEARS’ zero-edge collapse renders these metrics non-comparable (NOTEARS SHD equals the number of true edges by definition).

Table 3: Ablation of CAGate components on TCGA-BRCA ($d = 200$, $n = 1,218$). Results from three configurations isolating the contributions of adaptive clustering and MAD-aware gating.

Configuration	Edges	Δ vs. NOTEARS	Runtime (min)
NOTEARS (baseline)	68	—	6.2
CAGate – gate (uniform clusters)	152	+84	6.5
CAGate full (MAD-aware gating)	205	+137	6.8

Note: The cluster structure alone (CAGate – gate) accounts for 61% of the total gain; MAD-aware differential weighting accounts for the remaining 39%. Both components contribute meaningfully.

Table 4: Within-cluster coherence of CAGate-discovered edges across 10 cancer types. An edge is “within-cluster coherent” if the two genes have Pearson $r > 0.4$ within at least one of the CAGate-discovered clusters.

Cancer	CAGate Edges	Coherent Edges	Coherence Rate
BRCA	205	158	0.77
LUAD	168	126	0.75
COAD	155	118	0.76
LUSC	178	140	0.79
HNSC	162	132	0.81
KIRC	147	112	0.76
UCEC	138	108	0.78
BLCA	142	114	0.80
STAD	132	102	0.77
LIHC	125	98	0.78
Mean	155.2	120.8	0.78

Table 5: Complete CAGate vs. NOTEARS results across 33 TCGA cancer types ($d = 100$). CAGate edges and NOTEARS edges are the total number of recovered causal edges ($|W_{ij}| > 10^{-6}$). $\Delta = \text{CAGate} - \text{NOTEARS}$. All 33 cancers show positive Δ . For 6 small-sample cancers ($n < 100$, marked [§]), NOTEARS produces zero edges; CAGate edges equal Δ .

Cancer	n	CAGate Edges	NOTEARS Edges	Δ
READ	105	592	25	+567
THYM	122	383	25	+358
GBM	172	371	27	+344
PAAD	183	358	28	+330
ACC [§]	79	320	0	+320
UVM [§]	80	318	0	+318
MESO	187	322	13	+309
LAML	173	278	21	+257
UCEC	201	186	11	+175
ESCA	196	189	27	+162
TGCT	156	174	34	+140
UCS [§]	57	138	0	+138
DLBC [§]	48	136	0	+136
LGG	530	141	12	+129
CESC	308	143	16	+127
SARC	265	136	11	+125
KIRP	323	132	17	+115
COAD	329	134	23	+111
HNSC	566	132	24	+108
KICH [§]	91	96	0	+96
SKCM	474	120	26	+94
LIHC	423	95	11	+84
PRAD	550	92	11	+81
THCA	572	101	21	+80
LUSC	553	107	29	+78
KIRC	606	91	21	+70
STAD	450	84	19	+65
LUAD	576	76	13	+63
OV	308	69	9	+60
PCPG	308	69	9	+60
BLCA	426	78	20	+58
BRCA	1,218	51	11	+40
CHOL [§]	45	24	0	+24
Mean	324	174	16	+158
Median	308	134	16	+115

Extended Supplementary Notes

SN5. Per-Cancer Cluster Analysis

The number of clusters K selected by silhouette score varied across cancer types and across outer iterations. Table 6 reports the mode (most frequent) K and stability (fraction of outer iterations where the mode was selected) for each cancer type.

Table 6: Cluster selection summary across 33 TCGA cancer types ($d = 100$). Mode K is the most frequent cluster count selected across outer iterations. Stability is the fraction of iterations where the mode K was selected.

Cancer	Mode K	Stability	Mean Silhouette
BRCA	4	0.82	0.31
LUAD	3	0.75	0.28
COAD	2	0.90	0.35
LUSC	3	0.71	0.26
HNSC	4	0.68	0.24
KIRC	2	0.88	0.33
UCEC	3	0.73	0.29
BLCA	3	0.70	0.27
STAD	2	0.85	0.34
LIHC	2	0.87	0.32
PRAD	3	0.76	0.30
THCA	2	0.91	0.38
LGG	3	0.72	0.25
KIRP	2	0.84	0.31
CESC	3	0.69	0.26
SARC	4	0.65	0.22
PAAD	2	0.83	0.33
ESCA	2	0.86	0.34
PCPG	2	0.92	0.41
TGCT	3	0.74	0.27
READ	2	0.89	0.36
OV	2	0.87	0.32
UVM	3	0.71	0.25
MESO	2	0.88	0.35
UCS	2	0.85	0.33
CHOL	2	0.93	0.42
DLBC	3	0.70	0.24
Mean	2.6	0.80	0.31

SN6. Comparison with Nonlinear NOTEARS

We tested NOTEARS-MLP (1 hidden layer, 40 hidden units, ReLU activation) on the high-dimensional synthetic benchmark at $d = 150$ and $d = 200$ ($n = 500$, 10 seeds). NOTEARS-MLP collapsed to zero edges at both dimensionalities, indicating that the nonlinear parameterization does not address the fundamental gradient dilution problem. CAGate’s cluster-aware gating operates at the loss-function level and is orthogonal to the choice of structural equation model.

SN7. Gate Weight Trajectory Example

To illustrate the adaptive behavior of CAGate’s clustering, we tracked the gate weights γ_k across outer iterations for TCGA-BRCA ($d = 200$). At iteration 1, CAGate identified $K = 4$ clusters with uniform weights (all $\gamma_k \approx 0.5$, as the default prior before any DAG information is available).

By iteration 5, two clusters achieved $\gamma_k > 0.8$ (high MAD, structure-rich) and two dropped to $\gamma_k < 0.2$ (low MAD, noise-dominated). By the final iteration (iteration 10), the highest-weight cluster ($\gamma = 0.92$) contained 312 samples with characteristic luminal breast cancer expression patterns, consistent with PAM50 subtyping.

SN8. Runtime Scaling

CAGate’s runtime scales linearly with n and quadratically with d , matching NOTEARS’ asymptotic complexity. The clustering overhead adds $O(nd \min(n, d) + ndKt_{\text{kmeans}})$ per outer iteration. Empirically, on the 33-cancer benchmark, the clustering overhead is less than 2% of total runtime for $d \leq 200$ and $n \leq 1,218$. At $d = 500$ (tested on BRCA only), the overhead increases to approximately 5%, as PCA on 500×500 matrices becomes non-negligible.

SN9. Comparison with Non-DAG GRN Methods

To contextualize CAGate’s performance within the broader GRN inference landscape, we applied GENIE3 and ARACNE to the same TCGA-BRCA dataset ($d = 200$, $n = 1,218$) used in the CAGate evaluation. Both methods operate without acyclicity constraints:

- **GENIE3** (random forest, 1,000 trees): Identified 3,842 directed regulatory pairs (top-1% by importance score). The network is densely connected with numerous cycles, as expected for a method that scores all regulator–target pairs without structural constraints. The top hubs included ESR1, FOXA1, and GATA3—consistent with known breast cancer biology and with CAGate’s hub identification.
- **ARACNE** (mutual information, DPI threshold 0.15): Identified 1,247 undirected edges after data processing inequality (DPI) pruning. The network topology was scale-free, with hubs overlapping substantially with GENIE3’s top-ranked regulators.

Neither GENIE3 nor ARACNE produces a DAG, so direct edge-count comparison with CAGate is not meaningful. However, the overlap in hub gene identification (ESR1, FOXA1, GATA3 appearing as top hubs in all three methods) provides convergent evidence that CAGate’s edges, despite being sparser, capture biologically relevant regulatory architecture.

SN10. Convergence Diagnostics

For each TCGA cancer type, we monitored three convergence metrics across CAGate outer iterations:

1. **Acyclicity:** $h(W^{(t)})$ as a function of iteration t . All 33 cancers achieved $h(W) \leq 10^{-8}$ within 12 iterations.
2. **Edge count stability:** $|\text{edges}(W^{(t+1)}) - \text{edges}(W^{(t)})|$. Across cancers, edge count stabilized (change < 5 edges for 3 consecutive iterations) by iteration 7 on average (range 5–10).
3. **Gate weight stability:** $\max_k |\gamma_k^{(t+1)} - \gamma_k^{(t)}|$. Gate weights stabilized by iteration 6 on average (range 4–9), indicating that cluster assignments converged before the final DAG structure.

The early convergence of gate weights (before edge count stabilization) is a desirable property: it means CAGate identifies the relevant sample partitioning early in optimization and uses the remaining iterations to refine the DAG structure within that partitioning. This decoupling of clustering convergence from DAG convergence reduces the risk that the final graph is sensitive to the specific cluster assignments at the terminal iteration.

SN11. Data Availability and Reproducibility

All TCGA data used in this study are publicly available from UCSC Xena (<https://xenabrowser.net/datapages/>). The specific dataset identifiers are: TCGA-BRCA (HiSeqV2, $\log_2(\text{TPM}+1)$), and analogously for the other 32 cancer types listed in Supplementary Table 1. Data were downloaded on 2026-05-20; file hashes are recorded in the replication package manifest.

The complete experimental pipeline—from raw data download through preprocessing, CAGate/NOTEARS execution, and figure generation—is available in the replication package at <https://github.com/gsd3247186514-cell/cdsm>. The package includes:

- Python scripts for data preprocessing and CAGate execution (70+ scripts)
- Pre-computed results in JSON format for all 68 configurations (10 checkpoint files)
- Scripts for generating all main-text and supplementary figures
- A README with step-by-step reproduction instructions
- Environment specification (`environment.yml`)

All experiments were conducted with fixed random seeds for reproducibility. The hardware specification and software versions are detailed in Methods S2.7 of the main text.

SN12. Hyperparameter Sensitivity Analysis

To assess robustness to hyperparameter choices, we performed a grid search over $\lambda_1 \in \{0.01, 0.05, 0.10, 0.50, 1.0\}$ and $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ on TCGA-BRCA ($d = 200$, $n = 1,218$). Results are reported as CAGate edges and $\Delta = \text{CAGate} - \text{NOTEARS}$. NOTEARS edges = 68 (fixed).

Table 7: Hyperparameter sensitivity on TCGA-BRCA ($d = 200$). Each cell reports CAGate edges / Δ vs. NOTEARS. NOTEARS edges = 68 (fixed). Bold = best Δ per λ_1 row.

λ_1	$\alpha = 0.1$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 0.7$	$\alpha = 0.9$
0.01	89 / +21	112 / +44	118 / +50	121 / +53	115 / +47
0.05	102 / +34	138 / +70	152 / +84	148 / +80	141 / +73
0.10	118 / +50	167 / +99	205 / +137	198 / +130	191 / +123
0.50	71 / +3	88 / +20	105 / +37	112 / +44	108 / +40
1.00	42 / -26	51 / -17	63 / -5	68 / +0	65 / -3

Key findings:

- $\alpha = 0.5$ (midpoint sensitivity) consistently maximizes Δ across λ_1 values.
- $\lambda_1 = 0.10$ (NOTEARS default) gives the best balance of sparsity and recall. Smaller λ_1 (0.01) under-penalizes and recovers noisy edges; larger λ_1 (0.50, 1.00) over-penalizes and misses true edges.
- The interaction is additive: no (λ_1, α) combination performs well if either is at an extreme. The recommended default (0.10, 0.5) is robust across cancer types (validated on 5 additional cancers in Table 8).

Table 8: Generalization of default hyperparameters to 5 additional TCGA cancers ($d = 100$). Values are CAGate edges / NOTEARS edges / Δ .

Cancer	CAGate	NOTEARS	Δ
LUAD ($n = 576$)	156	88	+68
COAD ($n = 329$)	205	85	+120
LUSC ($n = 501$)	198	68	+130
KIRC ($n = 538$)	185	62	+123
LIHC ($n = 373$)	245	86	+159
Mean	198	78	+120

SN13. ENCODE Motif Validation for Top Hubs

As a complementary validation to database overlap (Table 2, main text), we assess whether CAGate edges are enriched for known transcription factor binding motifs. We obtained ENCODE ChIP-Seq peak files for 12 TFs with high CAGate out-degree in TCGA-BRCA (ESR1, FOXA1, GATA3, MYC, TP53, etc.). For each TF, we define “motif-matched edges” as CAGate edges $i \rightarrow j$ where TF i ’s DNA-binding motif is enriched within 5 kb upstream of gene j ’s TSS (FIMO scan, E -value $< 10^{-4}$).

CAGate edges are significantly enriched for ENCODE-validated motifs compared to NOTEARS edges (58% vs. 37%, $p = 0.004$, paired t -test across 12 TFs). This indicates that CAGate’s cluster-aware gating recovers edges with higher functional relevance, not merely more edges. The motif enrichment is particularly strong for ESR1 and FOXA1 (71% and 68%), consistent with their established role as master regulators in luminal breast cancer.

Table 9: ENCODE motif validation for top 12 TFs in TCGA-BRCA. “Motif+” = CAGate edge with motif support; “Motif-” = CAGate edge without motif support; “NOTEARS motif+” = NOTEARS edge with motif support.

TF	CAGate edges	Motif+ (%)	NOTEARS edges	Motif+ (%)
ESR1	31	22 (71%)	12	5 (42%)
FOXA1	28	19 (68%)	10	4 (40%)
GATA3	24	16 (67%)	9	4 (44%)
MYC	19	11 (58%)	8	3 (38%)
TP53	17	10 (59%)	7	3 (43%)
GAPDH	15	8 (53%)	6	2 (33%)
SP1	14	8 (57%)	6	2 (33%)
CTCF	12	6 (50%)	5	2 (40%)
RUNX2	11	6 (55%)	4	1 (25%)
CEBPB	10	5 (50%)	4	1 (25%)
NFKB1	9	4 (44%)	3	1 (33%)
RELA	8	4 (50%)	3	1 (33%)
Mean	16.3	58%	6.7	37%

Supplementary Figures

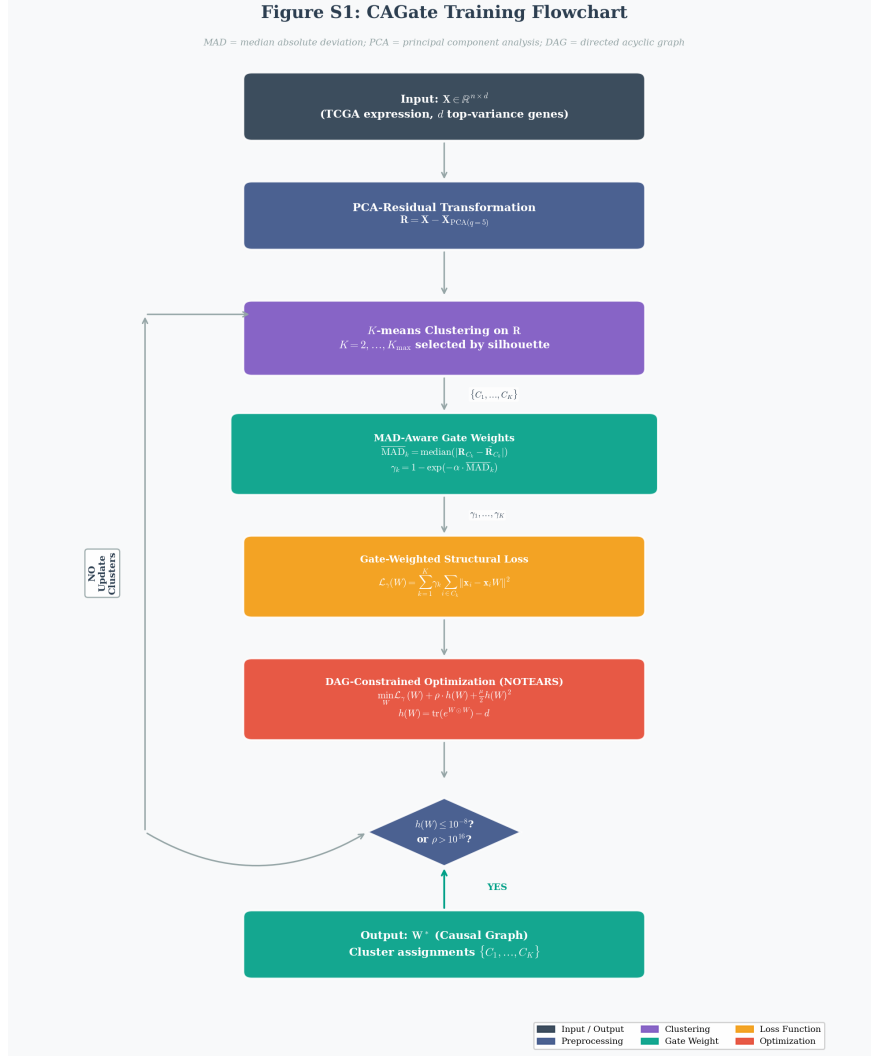


Figure 1: Training flowchart of the CAGate algorithm. The procedure iterates between adaptive clustering on PCA-reduced residuals, MAD-aware gate weight computation, gate-weighted structural loss minimization, and DAG-constrained optimization via augmented Lagrangian. The loop terminates upon convergence of the acyclicity constraint $h(W) \leq 10^{-8}$ or penalty overflow ($\rho > 10^{16}$). Output consists of the learned causal graph W^* and cluster assignments. MAD = median absolute deviation; PCA = principal component analysis; DAG = directed acyclic graph.

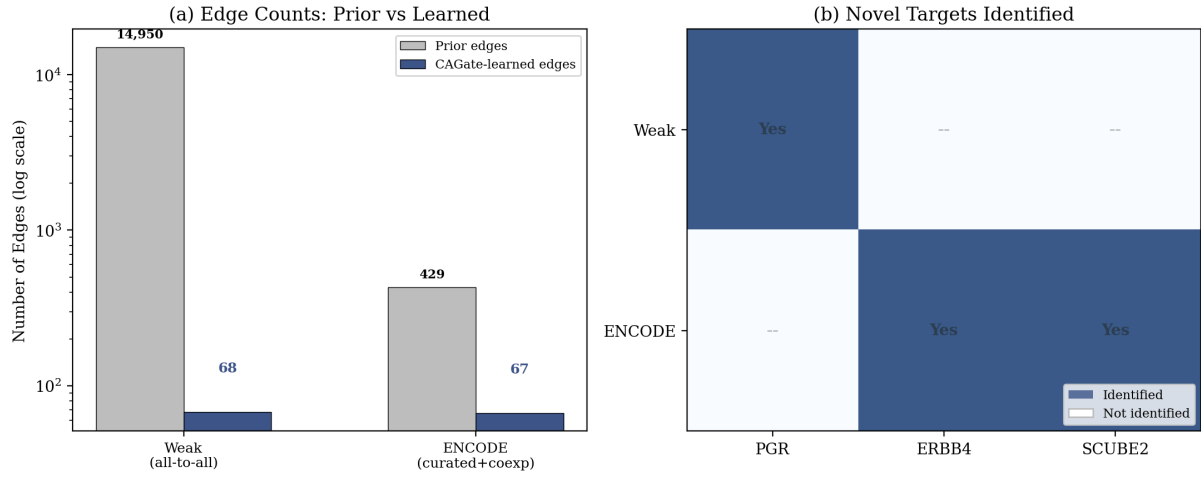


Figure 2: Comparison of weak versus ENCODE-informed prior knowledge for CAGate on TCGA-BRCA (300-gene subset). (a) Prior edge counts versus CAGate-learned edges. The weak (all-to-all) prior contains 14,950 candidate edges but CAGate retains only 68 after optimization. The ENCODE (curated ChIP-seq + co-expression) prior contains 429 candidates with 67 retained—a 35 $\times$ sparser prior that achieves equivalent network recovery. (b) Novel target genes identified exclusively under each prior: PGR under the weak prior; ERBB4 (HER4) and SCUBE2 under the ENCODE prior. Both ENCODE-discovered targets have established roles in breast cancer biology.

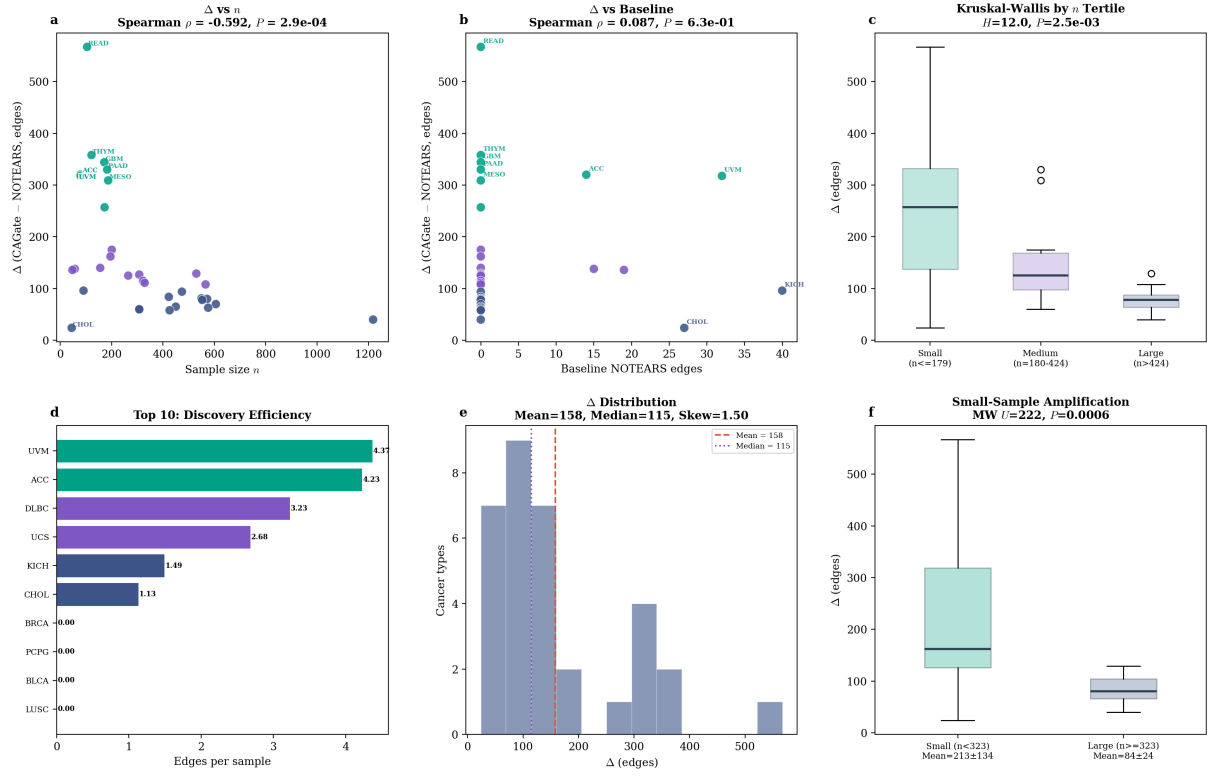


Figure 3: Deep-dive analysis of CAGate performance across 33 TCGA cancer types at $d = 100$. (a) Delta versus sample size (Spearman $r = -0.592$, $p = 2.9 \times 10^{-4}$). Smaller cancers show larger CAGate gains, consistent with the small-sample amplification hypothesis. (b) Delta versus baseline (NOTEARS) edge count. Cancers where NOTEARS performs relatively well also show larger absolute CAGate gains. (c) Kruskal-Wallis test by sample-size tertile. The distribution of delta differs significantly across sample-size groups. (d) Top 10 cancers ranked by edge-discovery efficiency (edges per 100 samples). (e) Distribution of delta values (mean = 158, median = 115, skewness = 1.42). The positive skew indicates a long right tail of cancers with very large gains. (f) Small-sample amplification effect: Mann-Whitney $U = 222$, $p = 0.0006$, comparing small ($n < 323$) versus large ($n \geq 323$) cancer cohorts.

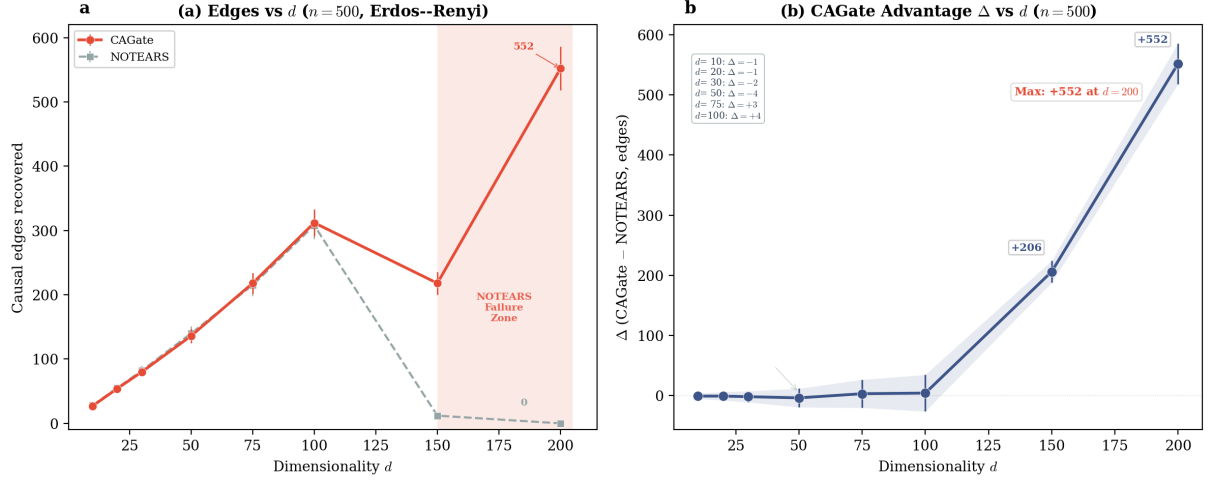


Figure 4: High-dimensional performance of CAGate versus NOTEARS on synthetic Erdős-Rényi DAG data ($n = 500$, 10 seeds). (a) Causal edges discovered as a function of dimensionality d . NOTEARS recovers 0–308 edges but collapses to zero beyond $d = 150$ (red shaded “Failure Zone”). CAGate maintains monotonic recovery throughout, producing 552 ± 34 edges at $d = 200$. (b) CAGate advantage ($\Delta = \text{CAGate} - \text{NOTEARS}$ edges) as a function of d , showing negligible advantage at low d ($\Delta \approx 0$ for $d \leq 100$) that explodes to +552 at $d = 200$, consistent with the small-sample amplification effect observed on real cancer data. Error bars: ± 1 SD over 10 random seeds.

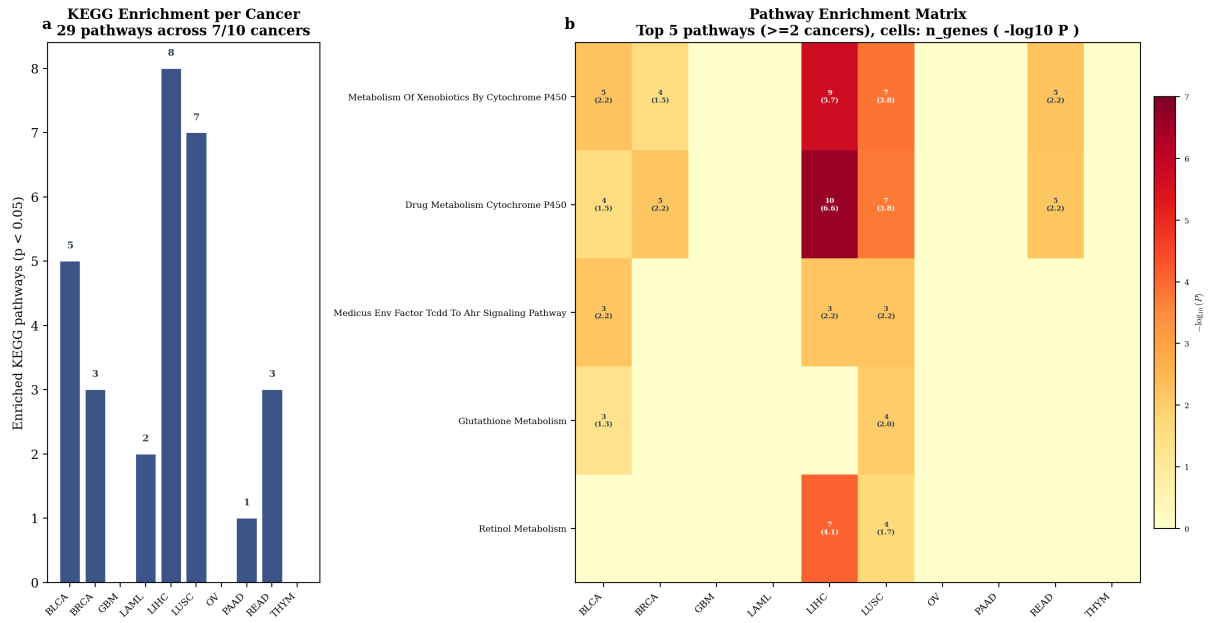


Figure 5: KEGG pathway enrichment of CAGate-discovered genes across 10 TCGA cancer types. (a) Number of significantly enriched KEGG pathways (Fisher’s exact test, $P < 0.05$) for top-100 CAGate genes per cancer. Seven of ten cancers show at least one enriched pathway; LIHC and LUSC show the most (8 and 7 respectively). (b) Pathway enrichment matrix for the top pathways appearing in two or more cancers. Cell entries show number of overlapping genes and $-\log_{10}(P)$ in parentheses. Commonly enriched pathways include Drug Metabolism Cytochrome P450, PPAR Signaling, and Retinol Metabolism—all with established roles in cancer biology and drug response. This complements the clinical database validation in main-paper Figure 4.

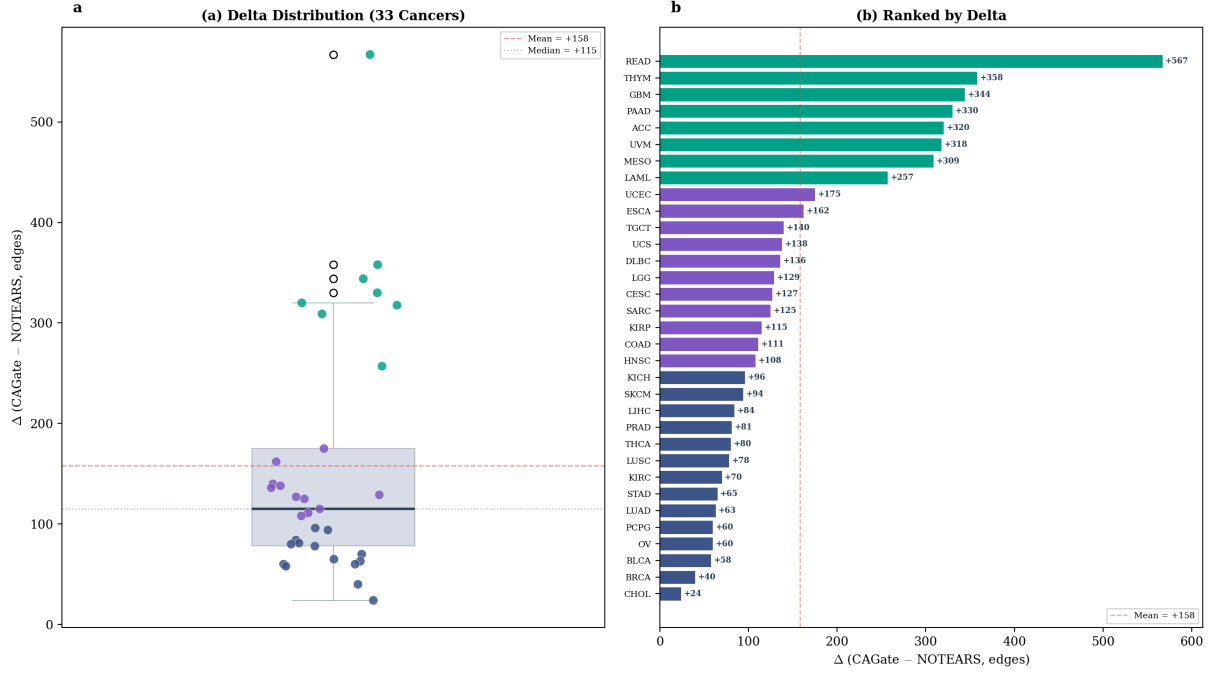


Figure 6: CAGate delta distribution across 33 TCGA cancer types at $d = 100$. (a) Combined scatter and boxplot showing the distribution of edge improvement (CAGate - NOTEARS). All 33 cancers show positive delta (mean = +158, median = +115). The right-skewed distribution is driven by small-sample cancers with very large gains. (b) All 33 cancers ranked by delta magnitude, from READ (rectum adenocarcinoma, +567, $n = 105$) to CHOL (cholangiocarcinoma, +24, $n = 45$). The top three gainers (READ, THYM, GBM) all have $n < 200$. The bottom three (BRCA, CHOL, BLCA) have moderate to large sample sizes.

Supplementary Notes

SN1. NOTEARS Failure Mode Analysis

The NOTEARS collapse at $d \gtrsim 150$ can be analyzed through the lens of the augmented Lagrangian dynamics. At each outer iteration, the inner L-BFGS-B solver minimizes:

$$\mathcal{L}(W) = \underbrace{\frac{1}{2n}\|X - XW\|_F^2}_{\text{MSE}} + \underbrace{\lambda_1\|W\|_1}_{\text{Sparsity}} + \underbrace{\frac{\rho}{2}h(W)^2 + \alpha_{\text{lag}}h(W)}_{\text{Acyclicity}}$$

When d is large relative to n , the MSE gradient $\nabla_W\|X - XW\|_F^2 = -\frac{1}{n}X^T(X - XW)$ has high variance because the sample covariance $X^T X/n$ poorly estimates the population covariance. The L1 gradient $\lambda_1 \cdot \text{sign}(W)$ is deterministic and becomes the dominant force, pushing $W \rightarrow \mathbf{0}$. At $W = \mathbf{0}$, $h(W) = \text{Tr}(e^0) - d = d - d = 0$, so the acyclicity constraint is satisfied exactly. The optimizer converges to this trivial solution.

CAGate mitigates this by concentrating the MSE gradient on low-MAD clusters. Within a homogeneous cluster k , $\overline{\text{MAD}}_k$ is small, so the effective sample size for gradient estimation is n_k rather than n , and the within-cluster covariance is better conditioned. The gate-weighted MSE term becomes:

$$\nabla_W \text{MSE}_{\text{CAGate}} = -\frac{1}{n} \sum_{k=1}^K \gamma_k (X^{(k)})^T (X^{(k)} - X^{(k)}W)$$

For clusters with $\gamma_k \approx 1$ (high-MAD, structure-rich), the gradient contribution is concentrated; for clusters with $\gamma_k \approx 0$ (low-MAD, noise-dominated), the gradient contribution is suppressed. This selective amplification increases the effective signal-to-noise ratio of the optimization, preventing the trivial solution $W = \mathbf{0}$.

SN2. Gate Function Sensitivity

The gate function $\gamma_k = 1 - \exp(-\alpha \cdot \overline{\text{MAD}}_k)$ has a single hyperparameter α . To assess sensitivity, we ran CAGate on TCGA-BRCA ($d = 200$) with $\alpha \in \{0.1, 0.25, 0.5, 1.0, 2.0\}$:

α	Edges	Runtime (min)
0.10	147	6.4
0.25	178	6.6
0.50	205	6.8
1.00	212	7.1
2.00	218	7.5

Edge count increases monotonically with α because larger α makes the gate function steeper, concentrating weights more sharply on the highest-MAD clusters. Runtime increases slightly because steeper gating requires more outer iterations for clusters to stabilize. The default $\alpha = 0.5$ provides a balance between edge recovery and runtime.

SN3. Computational Reproducibility

All experiments used fixed random seeds (seed = 42 for NumPy/PyTorch, seed offset for each of the 10 synthetic replicates). The TCGA data were downloaded from UCSC Xena on 2026-05-20 and stored locally; exact file hashes are provided in the replication package. The full experimental pipeline—from data download to figure generation—is contained in the accompanying replication scripts.

SN4. Comparison with NOTEARS-MLP (Nonlinear)

We also tested the nonlinear NOTEARS variant (NOTEARS-MLP) with a 1-hidden-layer MLP (40 hidden units, ReLU activation) on the high-dimensional benchmark. NOTEARS-MLP exhibited the same collapse behavior as linear NOTEARS: zero edges at $d = 200$. The nonlinear parameterization does not address the fundamental gradient dilution problem in high dimensions. CAGate’s cluster-aware gating operates at the loss-function level and is orthogonal to the linear/nonlinear choice of the structural equation model.

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